

CHRISTINE GENG

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EDUCATION

University of Michigan

M.S. in Bioinformatics

Selected Coursework: Machine Learning, Bioinformatics Algorithms, Signal Processing

Expected May 2026

Ann Arbor, MI

GPA: 4.0/4.0

University of Michigan | LSA Honors

B.S. in Statistics and Data Science

Selected Coursework: Algorithms, Machine Learning, Real Analysis, Data Mining, Databases

Probability Theory, Computer Vision, Regression Analysis, Survival Analysis

Sept 2021 – May 2025

Ann Arbor, MI

GPA: 3.6/4.0

EXPERIENCE

CVS Health

Machine Learning Graduate Intern

May 2025 – Aug 2025

Boston, MA

- Increased accuracy by 9% of 30-day hospital readmission risk XGBoost model by introducing new patient-notes feature generated with Gemini 1.5 Flash, developed in Python.
- Parallelized Gemini Flash inference in Python, achieving 42x faster processing for large-scale note analysis.
- Engineered end-to-end batch-inference pipeline on GCP using Vertex AI Pipelines, Cloud Storage, and BigQuery, enabling efficient processing of clinical notes for over 10 million patients.

Flatiron Health

Machine Learning Engineer Intern

May 2024 – Aug 2024

New York, NY

- Developed clinical document categorization service in Python, increasing accuracy to 95% using LLMs, K-means clustering, and active learning.
- Built Human-in-the-Loop training framework to streamline labeled data generation; integrated into patient manager system. Created native AWS backend in Python using DynamoDB, S3, Lambda, and Step Functions to orchestrate inference and labeling workflows.
- Analyzed categorization errors to identify optimal model improvements.

Biomedical & Clinical Informatics Lab, University of Michigan

Machine Learning Research Assistant

Jul 2023 – Jan 2025

Ann Arbor, MI

- Conducted ML research to detect traumatic brain injury in CT scans, reducing reliance on physical exams
- Developed U-Net CNN to identify axial CT slices containing swelling optic nerves
- Optimized candidate selection pipeline via open-source libraries, reducing runtime by 89.4%
- First author of *Automated Semi-Supervised Measurement of Optic Nerve Sheath Diameter from CT Following Traumatic Brain Injury*, publication currently under review.

EXTRACURRICULAR EXPERIENCE

Girls Who Code at University of Michigan

Executive Board Member

Sep 2022 – Present

Ann Arbor, MI

- Led outreach program teaching Python and CS fundamentals to underrepresented high school students
- Mentored capstone projects on data analysis, visualization, and hypothesis-driven research

The Dialectic: Multi-Agent LLM Debate System | Python, Next.js

- Built multi-agent debate framework with real-time updates for LLM agents (Speaker, Challenger, Judge)
- Implemented counter-example guided loops to improve reasoning accuracy and reduce bias
- Developed interactive Next.js UI supporting live debate visualization, configurable rounds, and topic categorization

SKILLS & HONORS

Languages: Python, C++, R, SQL, Java, MATLAB

Libraries/Frameworks: TensorFlow, PyTorch, Keras, fastai, scikit-image, OpenCV, pydicom

Tools/Platforms: Docker, AWS, GCP, Vertex AI, Anaconda

Honors: Sophomore Honors Award, University Honors, Scholastic Writing Award, Scholastic Art Award